

Notes: Almeida and Wolfenzon (JF, 2006)
A Theory of Pyramidal Ownership and Family Business Groups

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1 Big picture

- This paper builds a formal theory of why family business groups use **pyramidal ownership structures**. The central question is not simply why families want control, but why they control firms through chains of ownership rather than by directly holding shares or using dual-class shares.
- The standard explanation says that pyramids are devices for separating voting control from cash-flow rights. Almeida and Wolfenzon argue that this explanation is incomplete because many pyramids involve only small deviations from one share–one vote, and because dual-class shares can often achieve similar deviations.
- The authors propose a financing-and-payoff rationale: a pyramid lets the family use the full retained earnings of an already controlled firm to finance a new firm, while also sharing the new firm’s nondiverted payoff with minority shareholders of the original firm.
- The model connects pyramidal ownership to weak investor protection, high private benefits of control, internal funds, and the endogenous selection of low-security-benefit firms into pyramids.

1.1 One-sentence summary

This note summarizes Almeida and Wolfenzon (2006), which develops a theory in which pyramidal family business groups arise because pyramids give controlling families both a financing advantage and a payoff advantage when investor protection is weak and new firms have low pledgeable or security benefits relative to required investment.

2 Research question

- Why do controlling families organize firms in pyramidal business groups rather than through direct horizontal ownership or dual-class shares?
- Why do family business groups arise in the first place, especially in countries with weaker investor protection?
- What types of firms are selected into pyramidal structures, and what does this imply for observed ownership concentration, diversion, and firm value?

3 Incremental contribution of the paper

- **Formal theory of pyramidal ownership.** The paper provides a formal model of pyramids, whereas much of the prior discussion relied on the informal idea that pyramids separate control from cash-flow rights.
- **Distinguishes pyramids from dual-class shares.** The model explains why pyramids can be optimal even when dual-class shares could achieve the same deviation from one share–one vote. This is important because the separation-of-control explanation cannot distinguish these mechanisms.
- **Links ownership structure to internal capital markets.** The key benefit of a pyramid is that the controlling family can access all retained earnings of a firm it already controls, not just the family’s own pro rata share.

- **Explains business groups and pyramids jointly.** The paper does not merely assume business groups exist. It shows why wealthy controlling families may be able to set up new firms when outside entrepreneurs cannot obtain external finance.
- **Provides a selection-based interpretation of poor performance and high diversion in pyramids.** Pyramidal firms may have lower value and more diversion not because pyramids mechanically cause expropriation, but because low-security-benefit firms are selected into pyramids.
- **Generates testable implications.** The paper predicts that pyramids should be more common under weak investor protection, in capital-intensive or low-profitability projects, after good performance of existing group firms, and in settings where retained earnings relax financing constraints.

4 Institutional and conceptual background

4.1 Family business groups

- A family business group is a collection of legally distinct firms controlled by the same family.
- Such groups are common in East Asia, Western Europe, Latin America, and many emerging markets.
- The paper focuses on family-controlled groups rather than looser group forms such as keiretsu-style coordination through banks or manager councils.

4.2 Pyramidal ownership

- A pyramid is a chain of control. The family controls firm A, firm A controls firm B, firm B may control firm C, and so on.
- In a simple two-layer pyramid, the family owns a fraction α of firm A, and firm A owns a fraction β^P of firm B. The family's ultimate cash-flow stake in firm B is $\omega^P = \alpha\beta^P$.
- In a horizontal structure, the family directly controls firm B, with ultimate ownership $\omega^H = \beta^H$.

4.3 Traditional view versus this paper

- **Traditional view:** pyramids allow families to control large assets with small cash-flow stakes.
- **Problem:** if dual-class shares are allowed, direct ownership with inferior-vote shares can also separate control from cash-flow rights.
- **This paper's view:** pyramids are useful because they change who finances new firms and who shares the new firms' nondiverted payoff.

5 Model setup

5.1 Timing

1. At date 0, a family sets up firm A and keeps a fraction α of its shares.
2. At date 1, firm A generates retained earnings c . A new investment opportunity, firm B, appears.

3. Firm B requires investment i at date 1 and produces cash flow r at date 2, with $r > i$ before diversion costs.
4. The family chooses whether firm B is set up through a horizontal structure or through a pyramid.

5.2 Key assumptions

- **Imperfect investor protection.** The controlling family can divert a fraction d of firm B's cash flows into private benefits.
- **Costly diversion.** If the family diverts d , it incurs a cost $c(d, k)r$, where k is investor protection. Higher k means diversion is more costly.
- **External investors break even.** Outside investors price securities rationally, anticipating diversion.
- **Control is retained.** The paper abstracts from losing control by assuming the family can retain control under either structure, for example through voting arrangements.

5.3 Security benefits and private benefits

- Security benefits are the nondiverted cash flows paid to all shareholders.
- Private benefits are the diverted cash flows captured by the family, net of diversion costs.
- The net present value net of diversion costs is

$$NPV = r - i - c(d, k)r.$$

6 Horizontal structure

6.1 Definition

- The family sets up firm B directly and uses its own personal wealth αc from firm A.
- The family contributes internal funds R_I^H and raises external funds R_E^H by selling $1 - \beta^H$ shares of firm B.

6.2 Family payoff

The family payoff in the horizontal structure is

$$\alpha c - R_I^H + \beta^H (R_I^H + R_E^H - i + (1 - d)r) + (d - c(d, k))r.$$

Because outside investors break even, the family ultimately captures the full NPV of firm B. The payoff can be written as

$$U^H = \alpha c + NPV.$$

6.3 Economic intuition

- Under horizontal ownership, nonfamily shareholders of firm A have no claim on firm B.
- Therefore, if firm B has a positive nondiverted payoff, the family prefers to own it directly and keep that payoff.

- The limitation is financing: the family can use only its own share αc of firm A's retained earnings.

7 Pyramidal structure

7.1 Definition

- Firm A sets up firm B as a subsidiary.
- Firm A can use all retained earnings c to finance firm B.
- The family indirectly owns firm B through its stake in firm A.

7.2 Family payoff

The family payoff in a pyramid is

$$\alpha (c - R_I^P + \beta^P (R_I^P + R_E^P - i + (1 - d)r)) + (d - c(d, k))r.$$

After imposing investor break-even, the family payoff becomes

$$U^P = \alpha c + NPV - (1 - \alpha)[(1 - d)r - i].$$

7.3 Economic intuition

- The term $(1 - \alpha)[(1 - d)r - i]$ is the part of firm B's nondiverted NPV that accrues to nonfamily shareholders of firm A.
- If this term is positive, the family dislikes sharing it and prefers horizontal ownership.
- If this term is negative, the family benefits from sharing the loss with nonfamily shareholders and prefers the pyramid.
- The pyramid also relaxes financing constraints because firm A can use all retained earnings c , not only the family's share αc .

8 Core mechanism: payoff advantage and financing advantage

8.1 Payoff advantage

- If firm B's nondiverted NPV is negative, then minority shareholders of firm A bear part of that negative security payoff in a pyramid.
- The family still obtains private benefits from diversion.
- Therefore, high diversion makes pyramids attractive because the family shares low security benefits but retains control benefits.

8.2 Financing advantage

- External investors discount securities when they expect diversion.
- Internal funds are therefore valuable because they avoid costly external financing.

- In a horizontal structure, the family can use only αc .
- In a pyramid, firm A can use all c , so the structure can finance projects that a horizontal structure cannot finance.

9 Main theoretical results

9.1 Result 1: basic structure choice

- If firm B's nondiverted NPV, $(1 - \bar{d})r - i$, is positive, the family chooses horizontal ownership.
- If firm B's nondiverted NPV is negative and the pyramid is feasible, the family chooses the pyramid.
- If financing is infeasible even in the pyramid, firm B is not set up.

9.2 Result 2: comparative statics

Firm B is less likely to be owned through a pyramid when:

- firm B has higher revenues r ,
- firm B requires a smaller investment i ,
- investor protection k is stronger.

Thus pyramids are most likely for low-security-benefit, high-investment, weak-investor-protection projects.

9.3 Result 3: formation of family business groups

- The paper introduces an outside entrepreneur who may be more productive than the family but has no internal wealth.
- When investor protection is weak, the entrepreneur may be unable to raise funds despite being more efficient.
- The wealthy family can set up the new firm because it has retained earnings in existing group firms.
- Business groups are therefore more likely when external finance is difficult.

9.4 Result 4: groups created by financing advantage are pyramids

- When the family controls firm B because it has a financing advantage rather than because it is the most efficient owner, the business group is organized as a pyramid.
- Horizontal groups can arise when the family is the efficient owner, but financing-advantage groups naturally lead to pyramids.

9.5 Results 5–7: ex ante optimality and timing

- Initial shareholders of firm A may rationally anticipate future pyramiding.
- If they anticipate it, the price of firm A adjusts so that they break even ex ante.

- The family may not want to commit against future pyramids because pyramids can relax future financing constraints.
- Pyramids are created over time and are more likely after good performance by existing group firms, because retained earnings become available for new investments.

10 Endogenous ownership concentration and diversion

10.1 Why endogenous ownership matters

- The simple model treats diversion as costless up to a threshold, so ownership concentration does not affect diversion.
- The extended model allows diversion to be costly and therefore makes diversion depend on ultimate ownership.
- This allows the authors to explain why observed pyramidal firms have lower ultimate ownership and higher diversion.

10.2 Costly diversion

Assume

$$c(0, k) = 0, \quad c_d > 0, \quad c_{dd} > 0, \quad c_{dk} > 0.$$

The family chooses diversion according to

$$d(\omega, k) = \arg \max_d \omega(1 - d) + d - c(d, k),$$

where ω is the family's ultimate ownership in firm B.

The first-order condition is

$$c_d(d(\omega, k), k) = 1 - \omega.$$

Thus:

- higher ultimate ownership ω reduces diversion;
- stronger investor protection k reduces diversion.

10.3 Result 8: family maximizes ultimate ownership

- In both horizontal and pyramidal structures, the family tries to maximize ultimate ownership subject to raising enough funds.
- The reason is that higher ownership reduces diversion costs.
- Therefore, the model does not assume that families want to dilute ownership as much as possible. Instead, dilution is a financing necessity.

11 Extracted figure from the paper

11.1 Read: Figure 1

- The vertical axis measures the required investment to set up firm B.

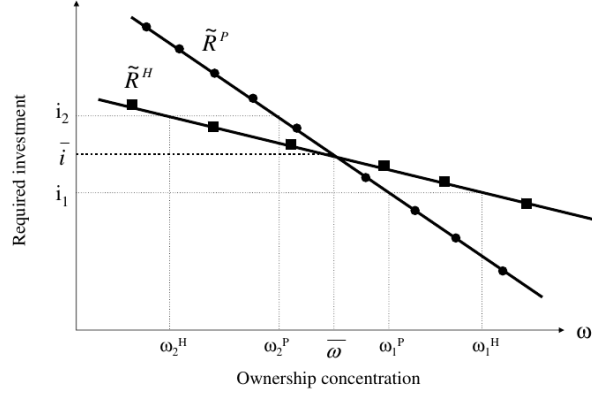


Figure 1. Determination of ultimate ownership concentration. Extracted directly from the paper.

- The horizontal axis measures the family's ultimate ownership concentration in firm B.
- The horizontal-structure financing curve is labeled $\tilde{R}^H$.
- The pyramidal-structure financing curve is labeled $\tilde{R}^P$.
- At the crossing point $\bar{\omega}$, the two structures raise the same funds for the same ultimate ownership concentration.

11.2 Economic message of Figure 1

- For low required investment i_1 , horizontal ownership supports a higher ultimate ownership concentration than pyramidal ownership. The family chooses the horizontal structure.
- For high required investment i_2 , the pyramid supports a higher ultimate ownership concentration than horizontal ownership. The family chooses the pyramid.
- The figure shows the selection logic behind the model: high-investment, low-security-benefit firms are selected into pyramids, while lower-investment or higher-security-benefit firms are more likely to be directly owned.

12 Observed ownership, diversion, and selection

12.1 Result 9: threshold logic

- There exists a threshold $\bar{\omega}$ at which the family is indifferent between structures.
- When investment requirements are high, revenues are low, or investor protection is weak, the pyramid is chosen.
- When investment requirements are low, revenues are high, or investor protection is strong, horizontal ownership is chosen.

12.2 Result 10: pyramids have lower observed ultimate ownership

- The model predicts that observed ultimate ownership in pyramidal firms is lower than in horizontal firms.

- But this is not because the family mechanically wants to separate cash flow from control.
- It is because firms requiring more external finance, and thus more dilution, are selected into pyramidal structures.

12.3 Result 11: investor protection and ownership stakes

- The ownership threshold below which the pyramid is chosen is decreasing in investor protection.
- In weak investor protection countries, pyramids can arise even with relatively high family cash-flow stakes.
- This explains why some pyramids have only small deviations from one share–one vote.

13 Extensions

13.1 Why firm B may be a separate legal entity

- A pyramid means firm B is legally separate from firm A, not merely a division.
- The paper discusses several reasons for separate incorporation: bankruptcy separation, managerial incentives, transparency, and the ability to sell claims directly on firm B.
- When investor protection or transparency differs across firms, it can be optimal to sell shares of firm B directly rather than raise funds only through firm A.

13.2 Variable investment scale and overinvestment

- The paper extends the model to allow the family to choose the scale of investment in firm B.
- If pyramidal ownership is chosen, the family may overinvest relative to the first-best scale.
- The reason is that the cost of overinvestment is partly borne by minority shareholders of firm A, while the family captures private benefits.
- This is a pyramidal version of Jensen’s free-cash-flow problem.

14 Empirical implications

14.1 Implications especially distinctive to this theory

1. **Pyramids may have high family cash-flow stakes.** Small separation between ownership and control does not refute the theory, because pyramids are not used only for separation.
2. **Pyramids can be chosen even when dual-class shares are available.** The pyramid has financing and payoff advantages that dual-class shares do not replicate.
3. **Pyramids are created over time.** They should often appear after good performance by existing group firms because retained earnings finance expansion.
4. **Pyramid subsidiaries are more likely to be separate-industry ventures.** If separate industries create different investor protection or transparency conditions, separate listing becomes more valuable.

14.2 Implications shared with the traditional view, but reinterpreted

1. **Diversion is higher in pyramidal firms.** In this paper, that occurs through selection: high-diversion firms are placed in pyramids.
2. **Pyramidal firms have lower value or performance.** Again, this can arise because low-profitability or high-investment projects are selected into pyramids.
3. **Family business groups are more common in weak investor protection countries.** Poor investor protection makes external finance difficult and increases the value of internal funds.
4. **Pyramidal firms are larger or more capital intensive.** High investment requirements make the pyramid's financing advantage more useful.
5. **Existing minority shareholders may lose when a new firm is added to the pyramid.** The pyramid can force them to share a negative nondiverted NPV.

15 Relation to the literature

15.1 Corporate governance and private benefits of control

- The paper builds on the literature showing that private benefits of control are larger when investor protection is weak.
- It is closely related to theories of tunneling, expropriation, and the valuation discount of firms with controlling shareholders.

15.2 Business groups and internal capital markets

- The model complements the view that business groups substitute for missing markets.
- It gives a sharper corporate-finance mechanism: the group exists because retained earnings in established firms relax external financing constraints.

15.3 Ownership structure and control mechanisms

- The paper contributes by comparing pyramids with horizontal ownership and dual-class shares.
- This distinction is central because many empirical studies observe pyramids in settings where dual-class shares could also be used.

16 How to read the model intuitively

16.1 Step 1: Think of firm B as a new project

- If firm B has high pledgeable income, outside investors are willing to finance it.
- In that case, the family prefers direct ownership because it keeps the benefits.

16.2 Step 2: Think of diversion as reducing pledgeable income

- Weak investor protection increases expected diversion.
- More diversion means less cash flow can be pledged to outside investors.
- Low pledgeable income makes external finance expensive or impossible.

16.3 Step 3: The pyramid solves financing but shifts costs

- Firm A's retained earnings finance firm B.
- Nonfamily shareholders of firm A share the nondiverted payoff of firm B.
- If the nondiverted payoff is low or negative, the pyramid is attractive to the family.

17 Strengths of the paper

- It explains why pyramids are distinct from dual-class shares.
- It links ownership structure to financing constraints and internal capital markets.
- It provides a selection-based interpretation of empirical patterns in pyramidal firms.
- It generates multiple comparative statics that can be taken to data.
- It bridges corporate governance, financial development, and family business group literatures.

18 Limitations and open questions

- The model is theoretical and stylized; real business groups often contain complex cross-holdings, multiple layers, and regulatory constraints.
- The model assumes that the family retains control under both structures, abstracting from explicit control contests.
- The welfare implications are not fully resolved. Pyramids may be privately optimal for the family but socially inefficient if they misallocate control or impede financial development.
- Empirical tests need to distinguish selection effects from causal effects of pyramidal ownership on expropriation and firm value.

19 Conclusion

19.1 Core conclusion

- Pyramidal ownership is not merely a mechanism to separate votes from cash-flow rights.
- A pyramid can be optimal because it allows the family to use all retained earnings of an existing controlled firm and to share low nondiverted payoffs with minority shareholders.

19.2 Economic intuition

- Weak investor protection reduces pledgeable income.
- Low pledgeable income makes external finance difficult.
- Established family firms have internal funds.
- A pyramid channels those internal funds into new firms while preserving family control.

19.3 Incremental contribution revisited

- The paper's central contribution is to reframe pyramids as a financing and payoff-sharing technology, rather than purely a vote-control technology.
- This reframing explains why pyramids appear even when cash-flow-vote deviations are small and when dual-class shares are available.

19.4 Takeaway

Pyramids are most useful when the controlling family wants to finance a new firm whose pledgeable security benefits are low: the pyramid uses internal funds, shares low security payoffs with minority shareholders, and lets the family retain control benefits.